

Inferring the hidden

Rebecca Akeresola

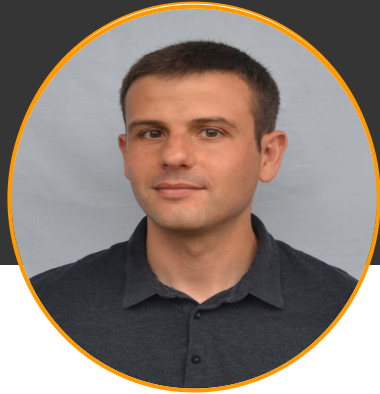
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BioSS Annual General Meeting

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PhD Supervisory Team



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Prof. Ruth
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Dr Gail
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Dr Adam
Butler





Previous
work



ongoing
work





Previous
work

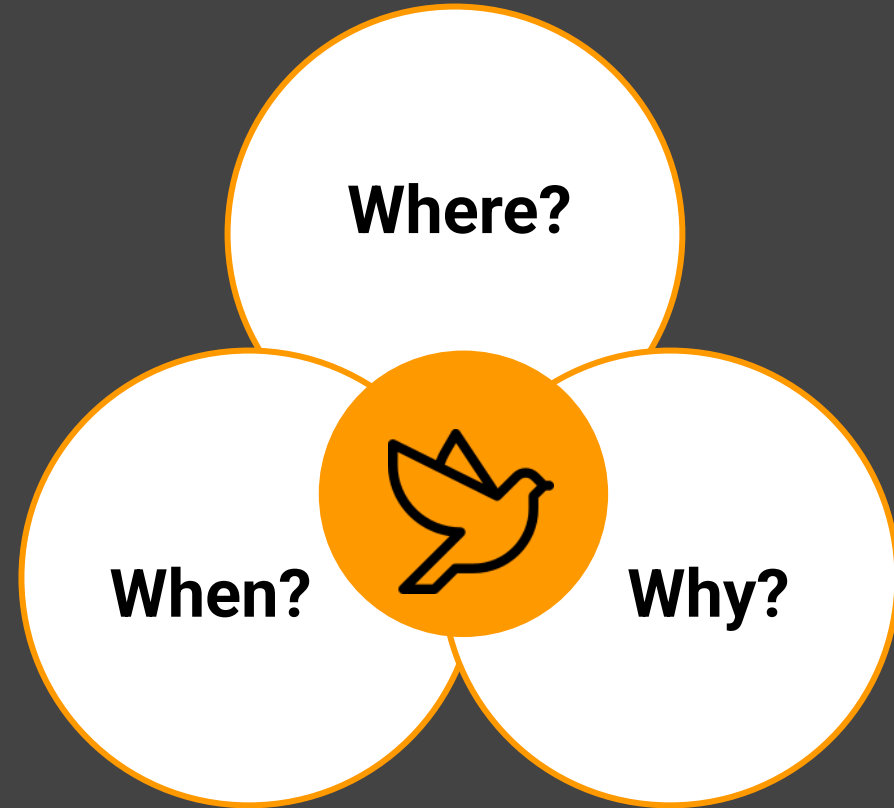


Previous
work

- Background

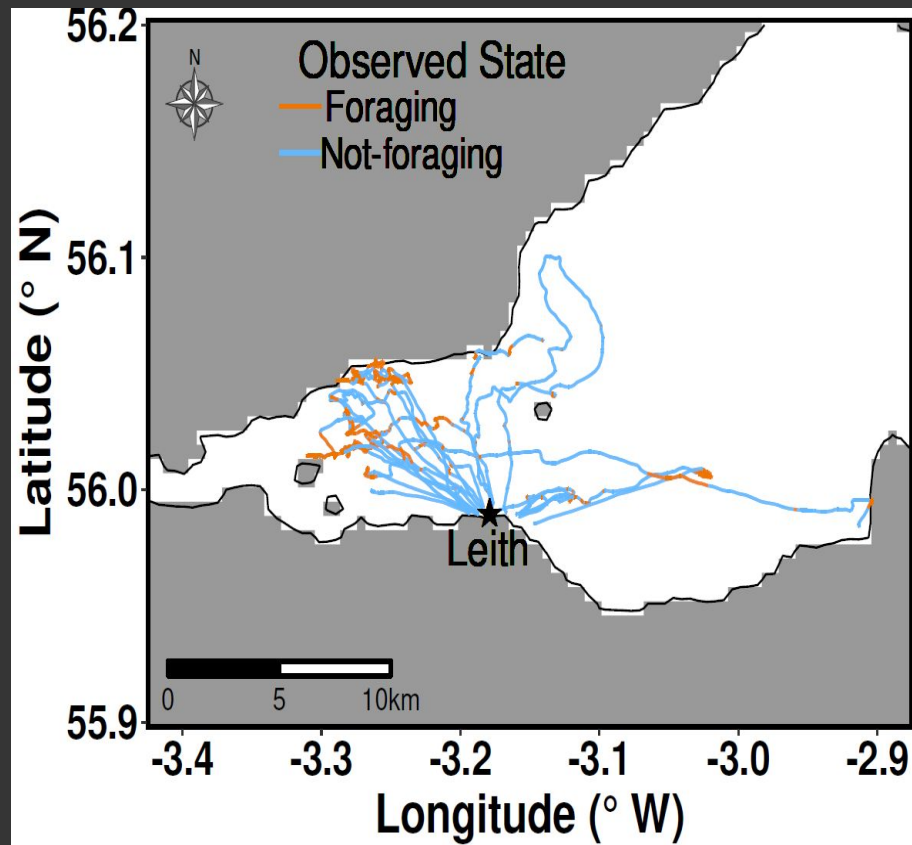


– Background



— Problem

- Behaviours inferred from movement data are rarely validated in practice
- Loss of useful information for analysis due to



— Goal



Validate inferred behaviours for effective conservation management

— Data



observed behavioural data

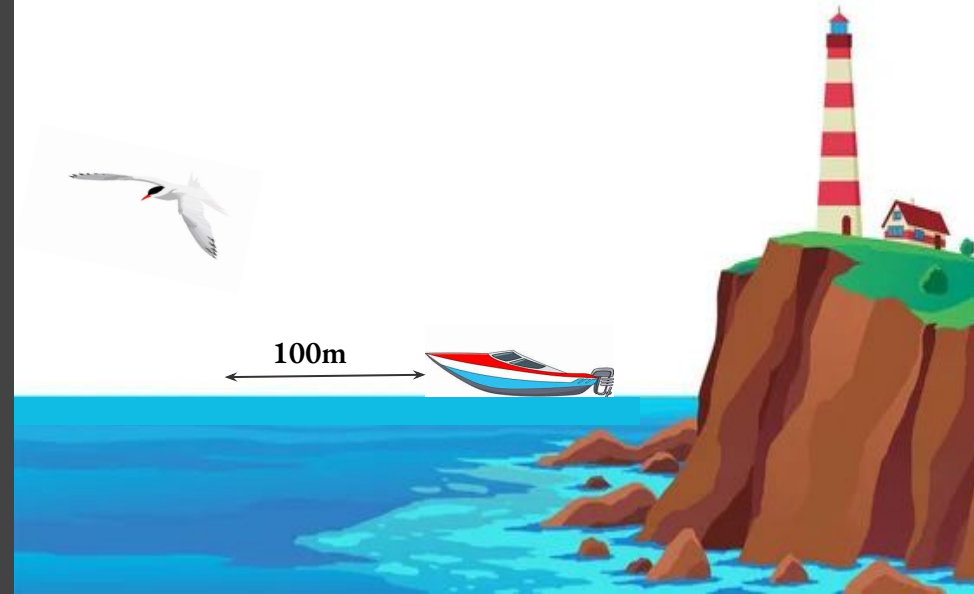


boat GPS location data

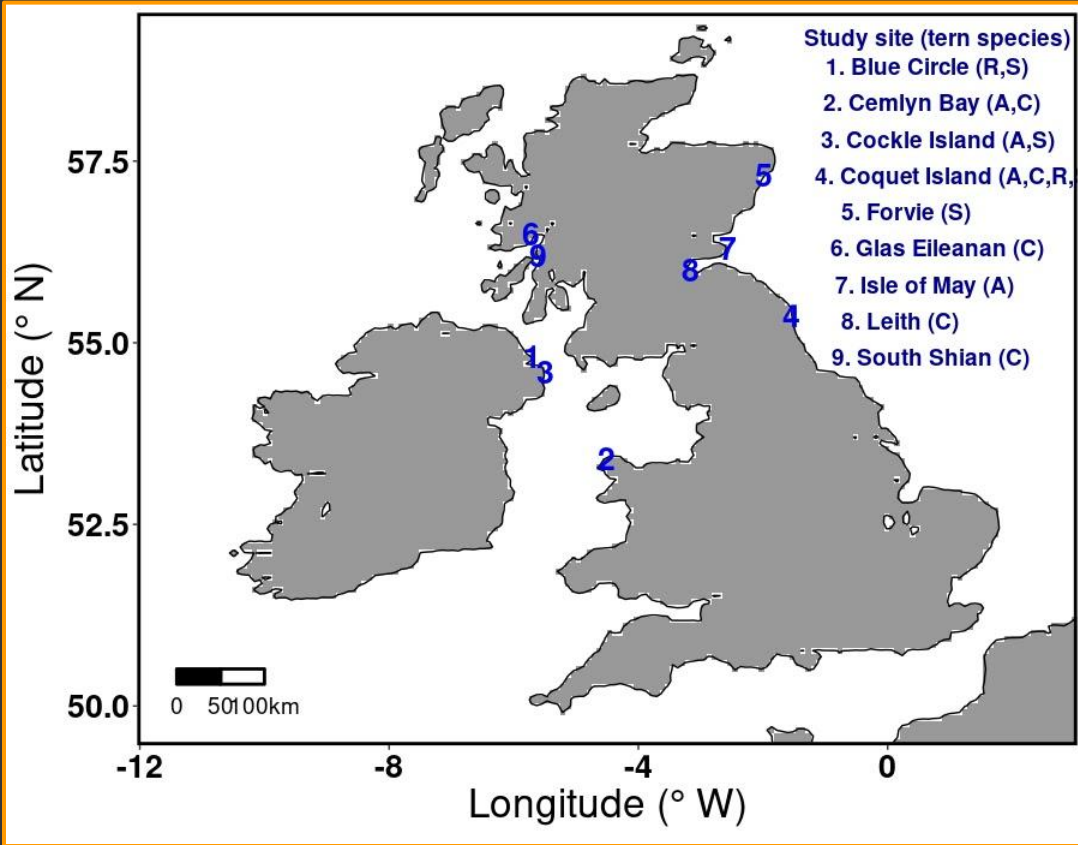
Sampling interval ~ 1-second

Method: visual tracking¹

Year: 2009-2011



Study sites



Arctic, A



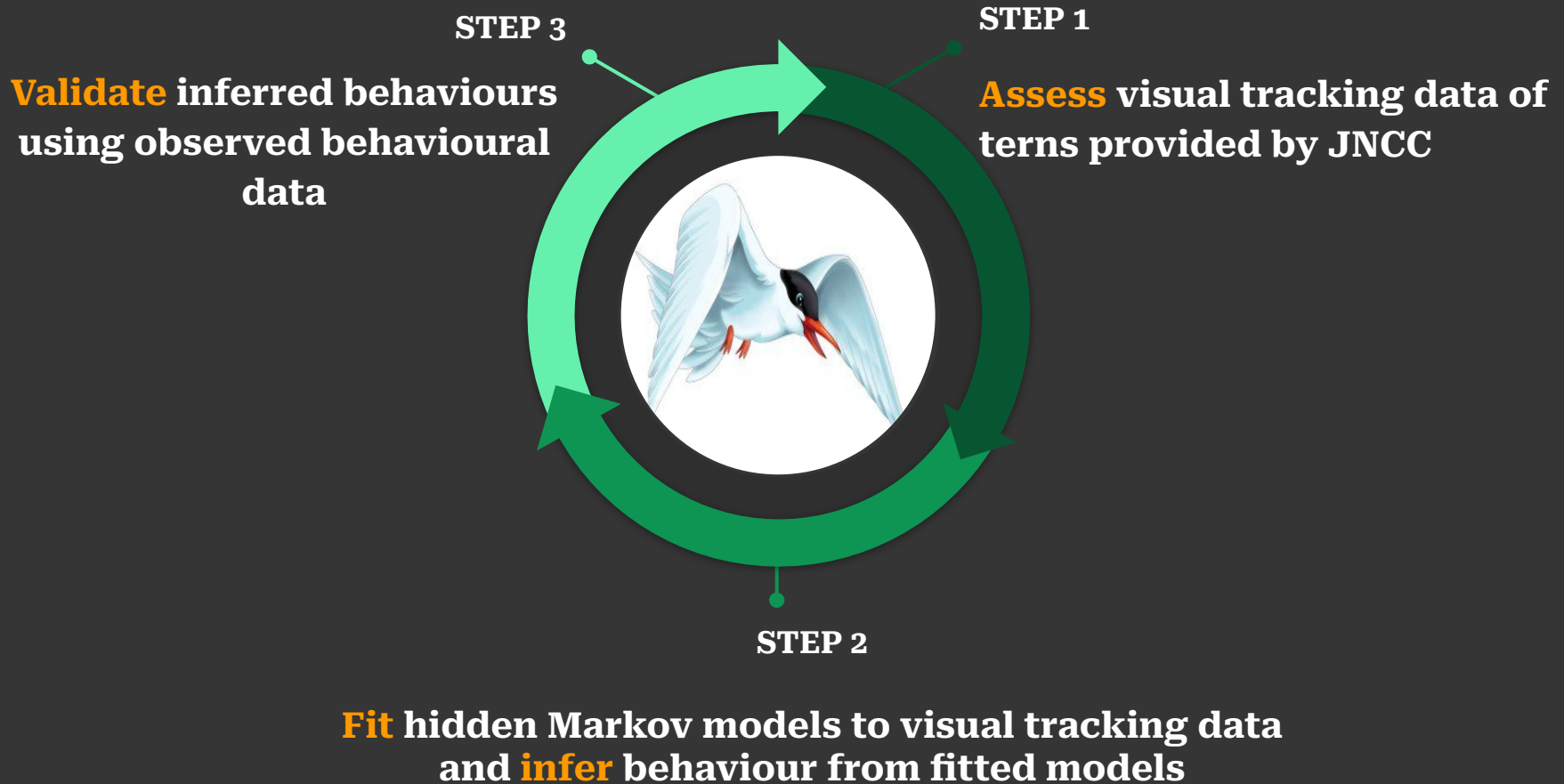
Roseate, R

Common, C



Sandwich, S

— Workflow

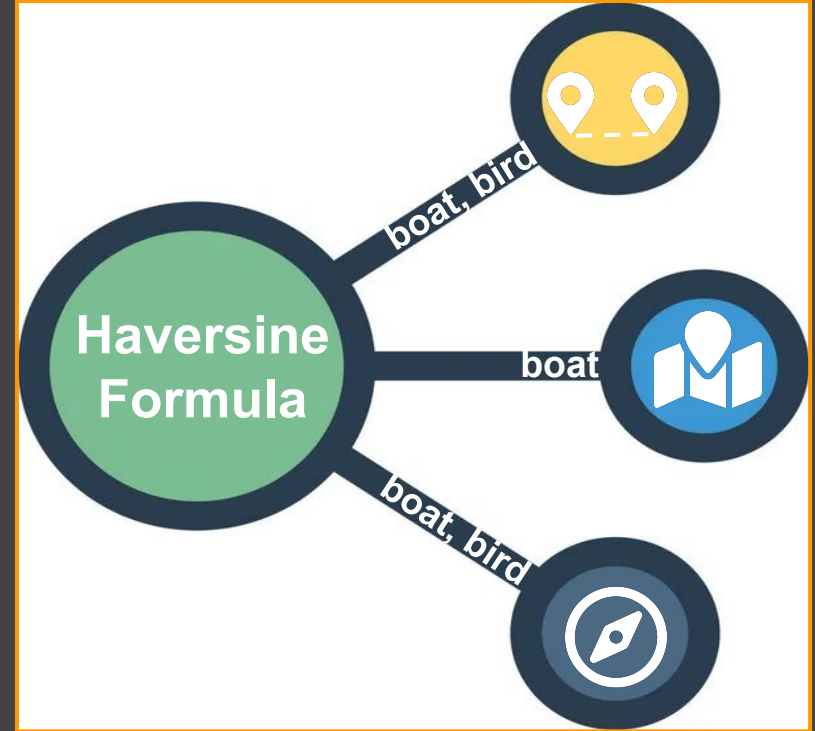
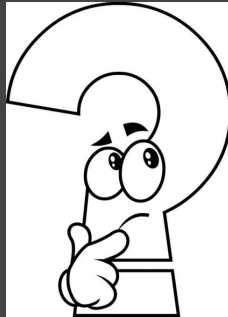
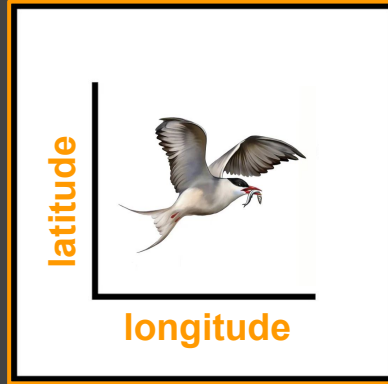
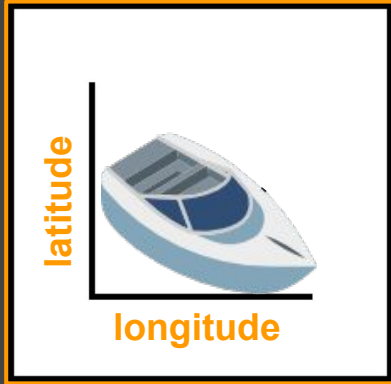


STEP 1

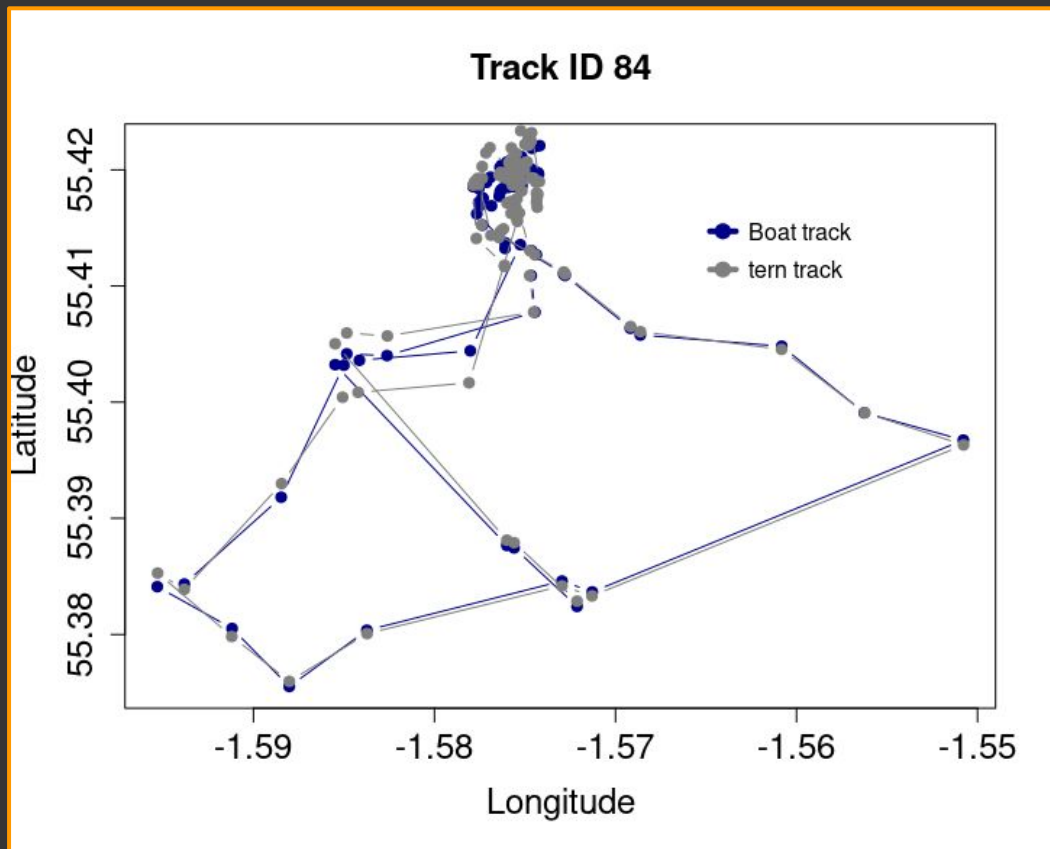
Assess visual tracking data of
terns provided by JNCC



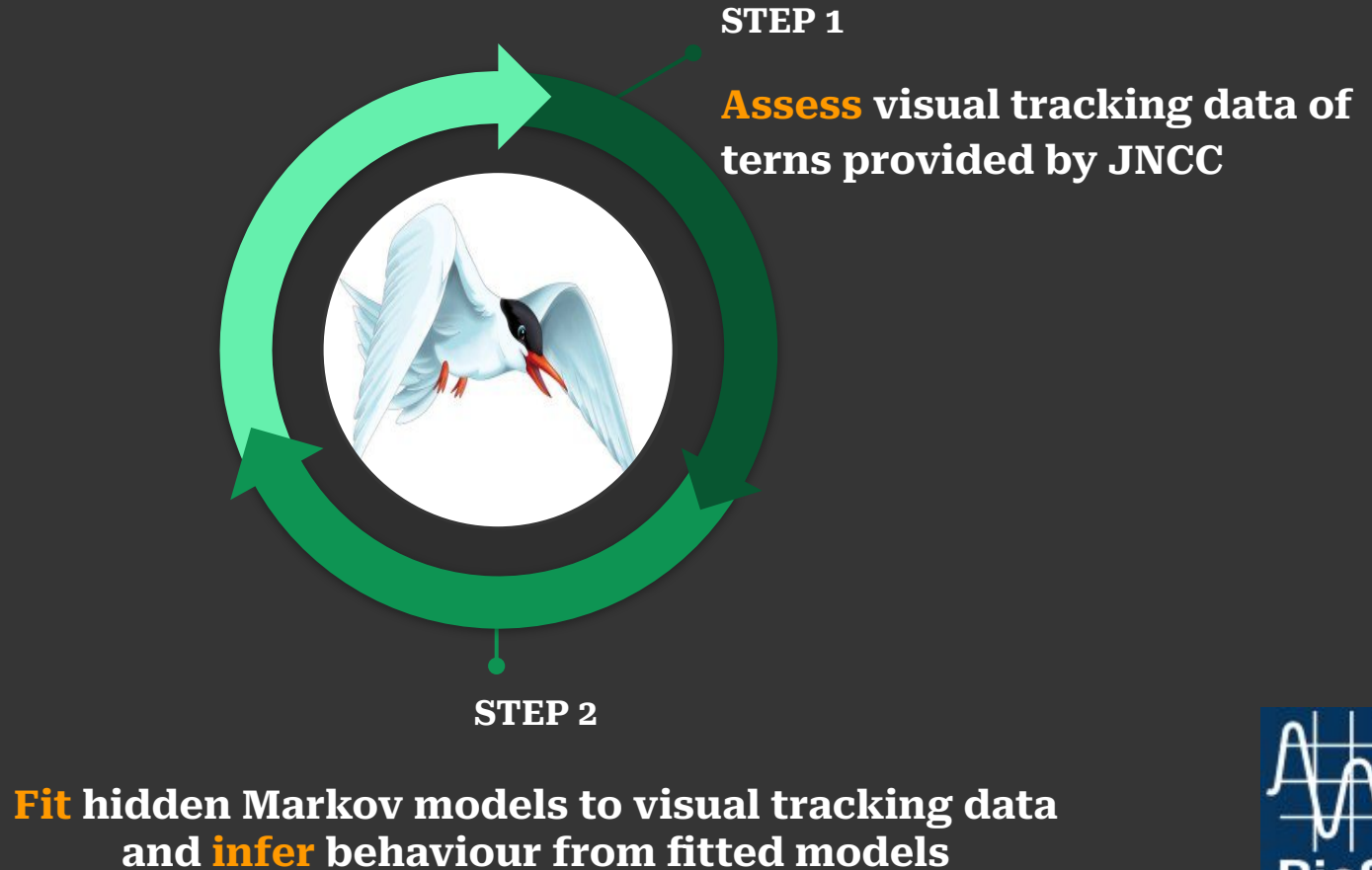
Assessing boat location data



Assessing boat location data

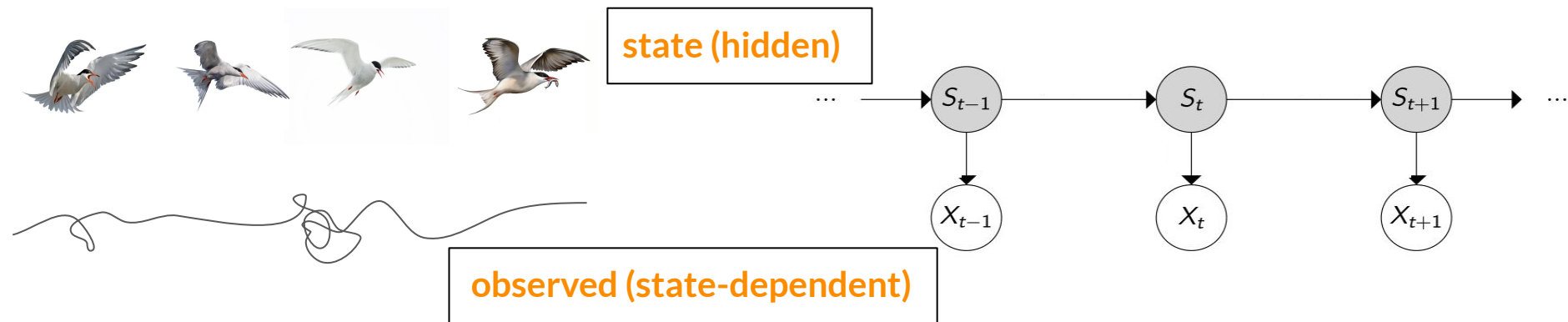


— Workflow



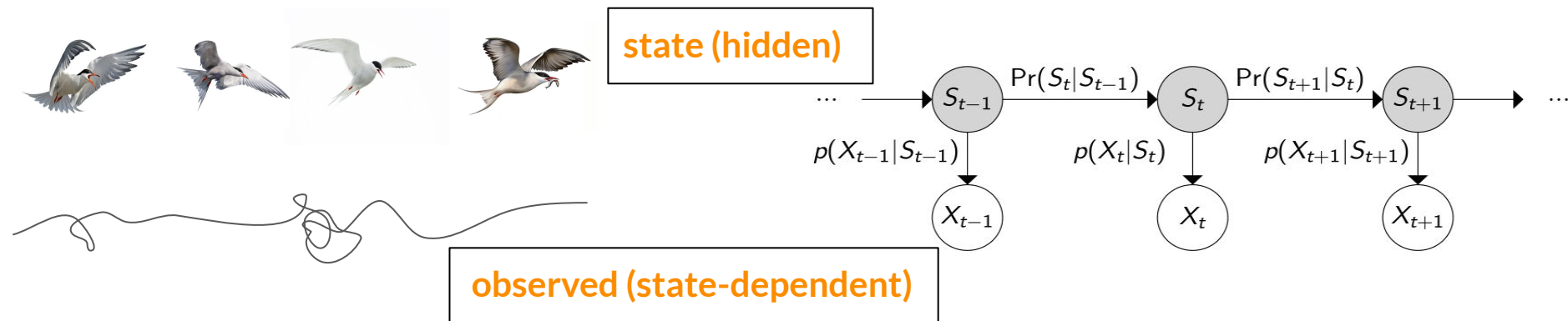
Hidden Markov model (HMM)

A time series model made up of two processes



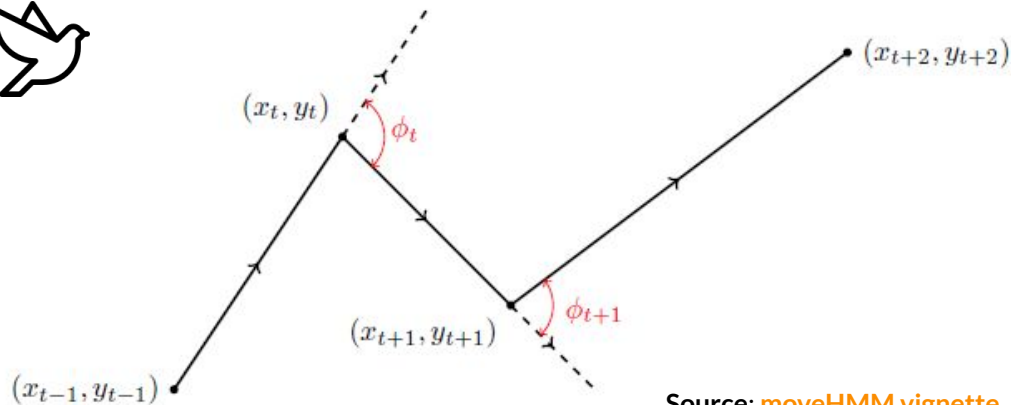
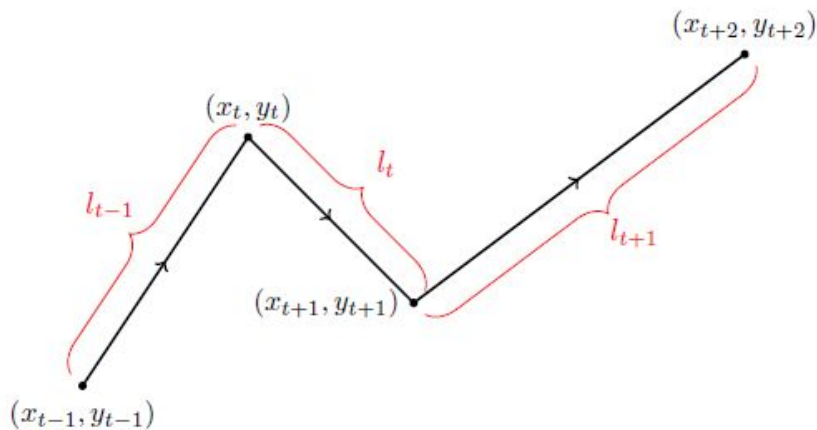
– We assume relationship **(Markov)** between

1. hidden sequence of states ($S_{\text{current}}|S_{\text{previous}}$)
2. observed and hidden states ($X_{\text{current}}|S_{\text{current}}$)



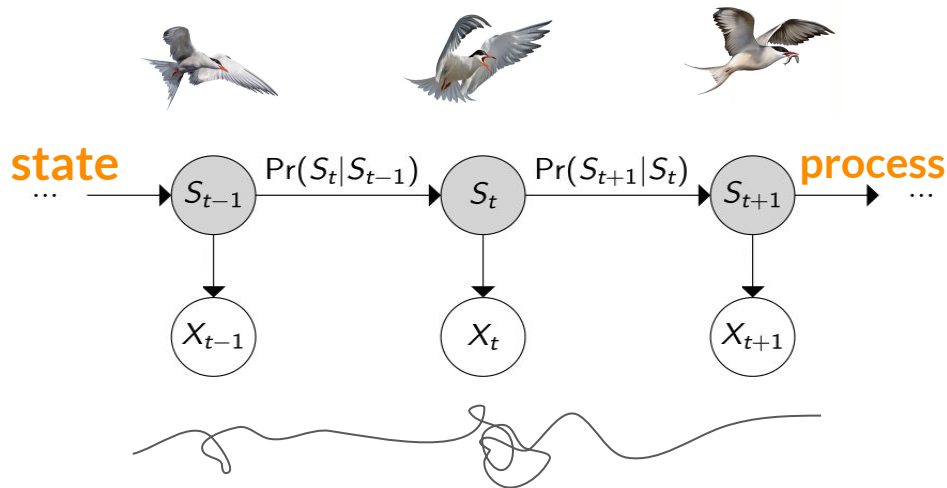
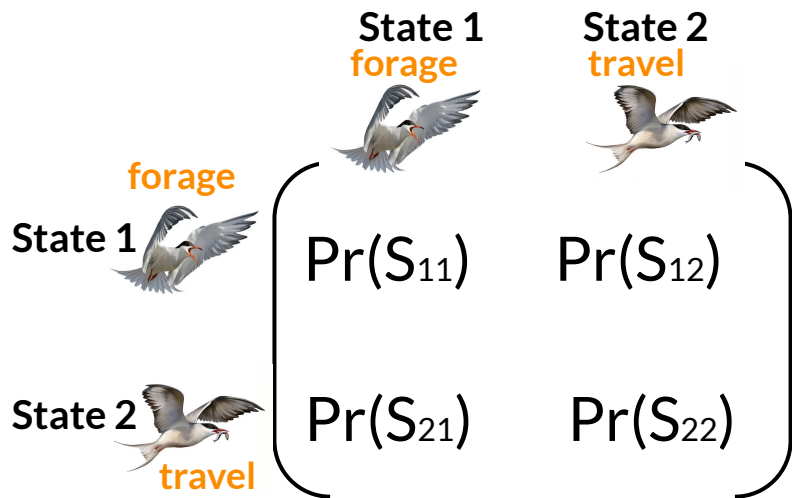
– HMM ~ Observed process

Specified via the
distributions of step length, l , and turning angles, ϕ
obtained from locations (x,y)

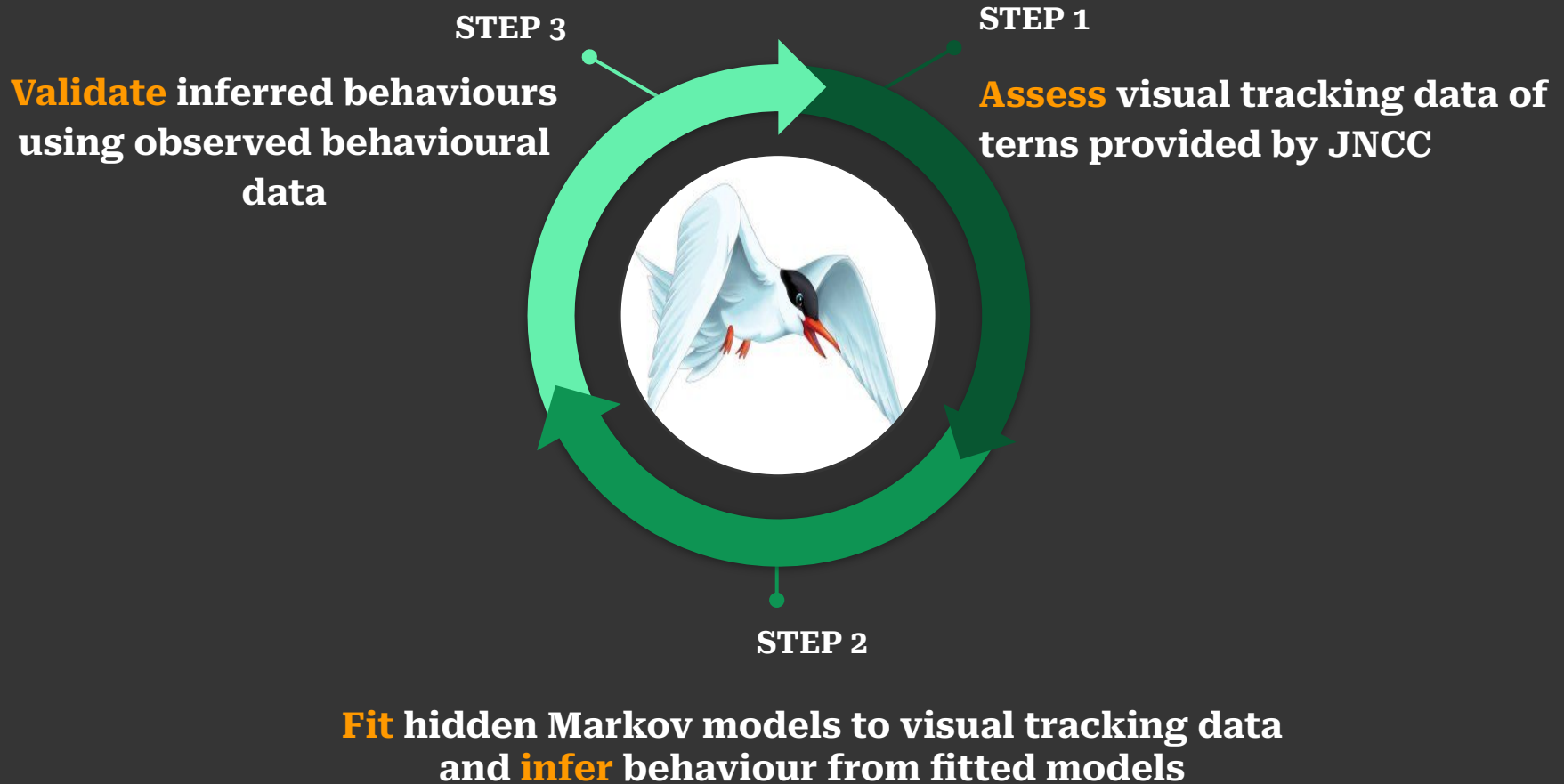


– Hidden Markov model ~ **State (hidden) process**

Specified via the **transition probability matrix**



— Workflow



Validation results

- HMMs performance in decoding behavioural states
- Correctly decoded foraging behavioural state during chick-rearing



Arctic tern

64-74%

58-70%

3 colonies



Common tern

76-84%

70-90%

5 colonies



Roseate tern

76%

86%

1 colony



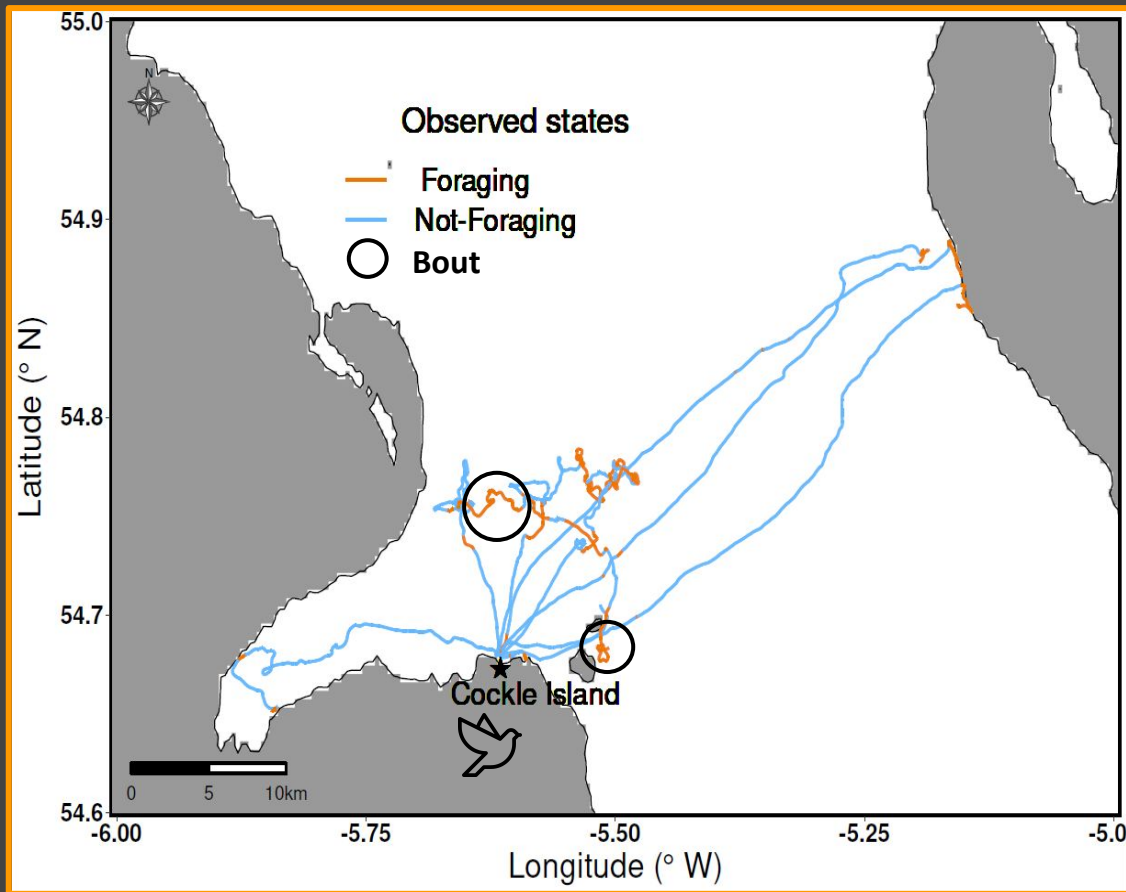
Sandwich tern

72-88%

74-91%

4 colonies

— Foraging event



A bout where foraging behavioural activity occur within the tracking data

Result: missed foraging event

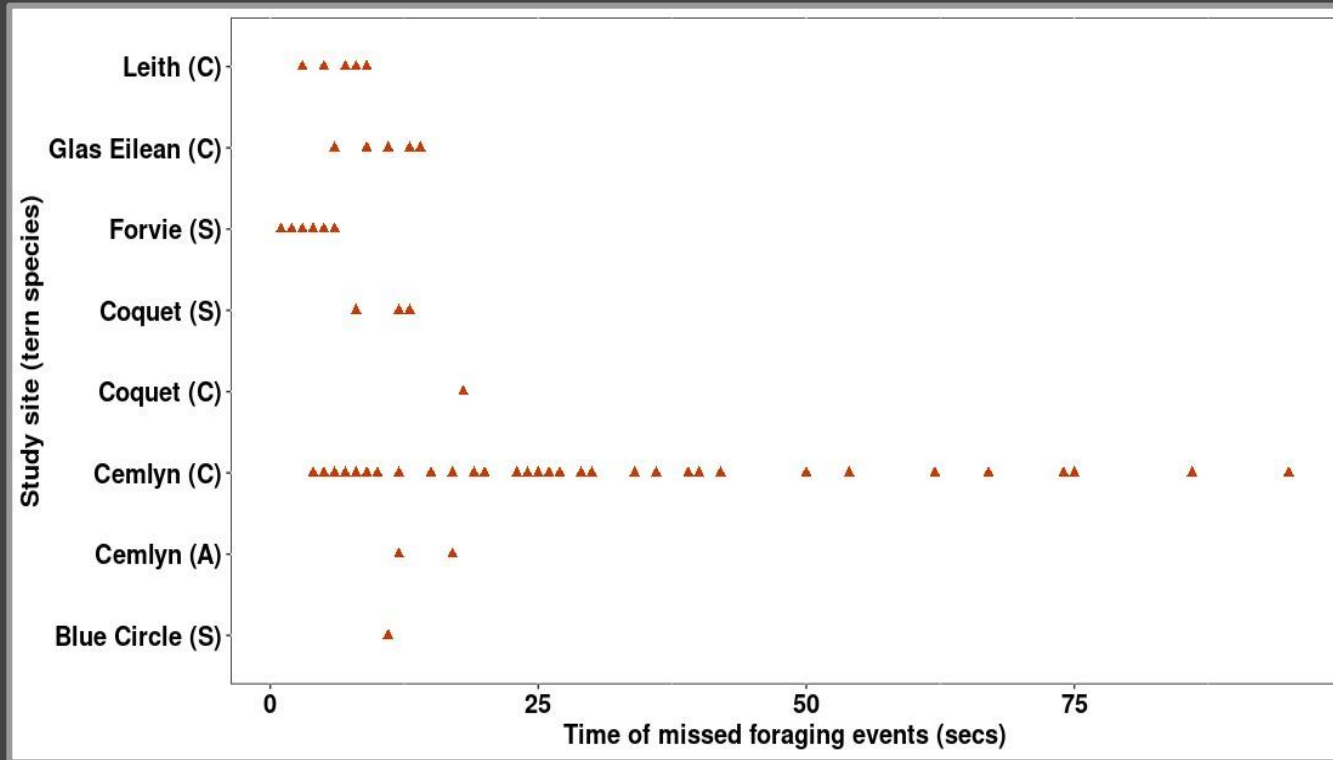


Figure: Observed foraging event completely missed from decoded behavioural states across study sites. A, C, S = Arctic, common and Sandwich tern

— Implication of findings

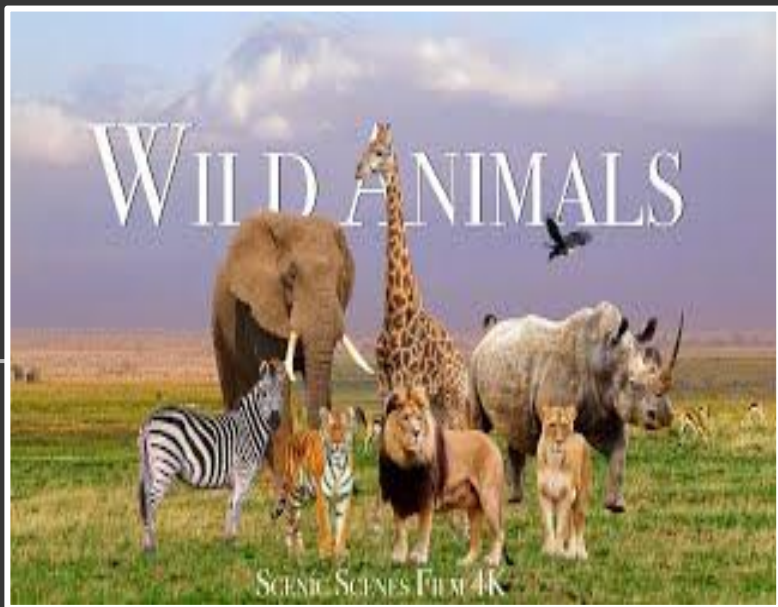


- Increased confidence in unvalidated inferred behaviours from telemetry data
- Conservation-relevant behaviours inferred using HMMs can inform the planning and design of designated protected areas

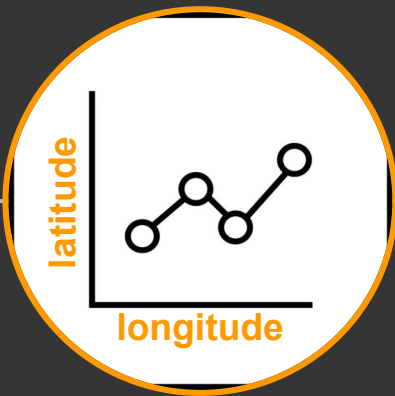


Ongoing
work

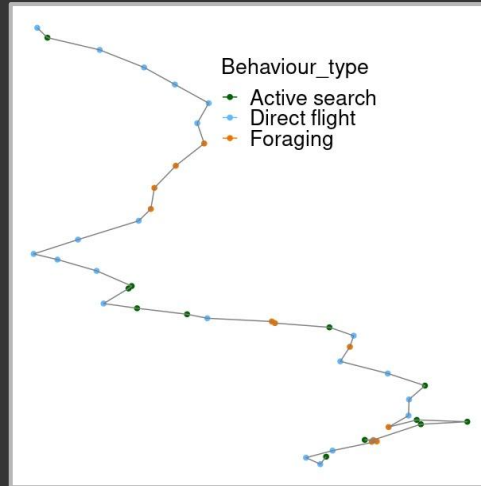
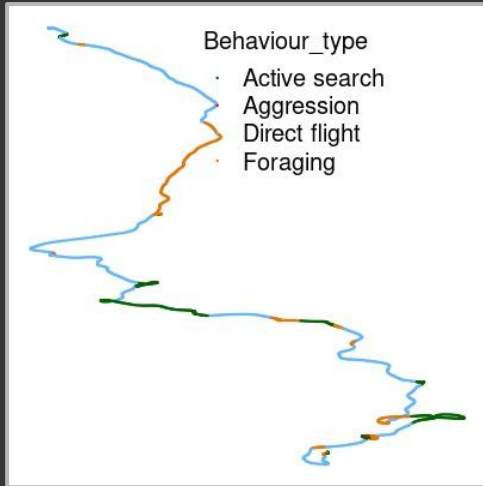
Background



Continuous
time



— Problem



How do we observe movements
at discrete-time with little or no
loss in underlying behaviour
influencing movement

Fine temporal scale

GPS battery life
Sampling duration

Coarse temporal scale

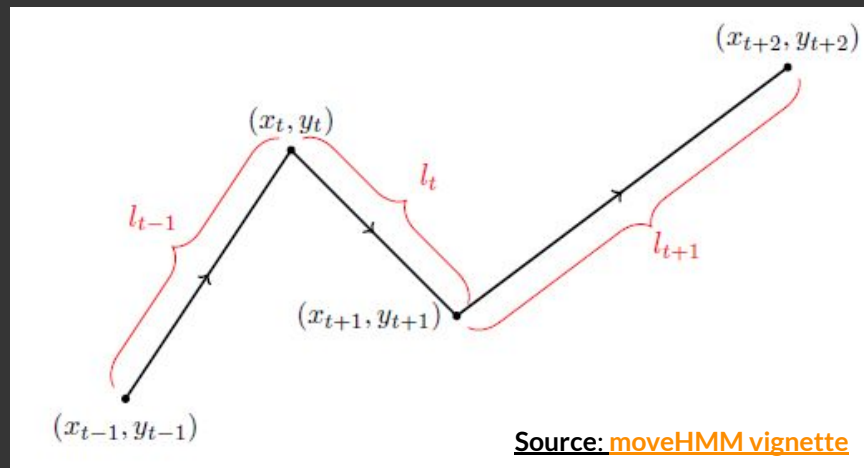
Miss out on event
obtainable at a fine-scale



— Goal?

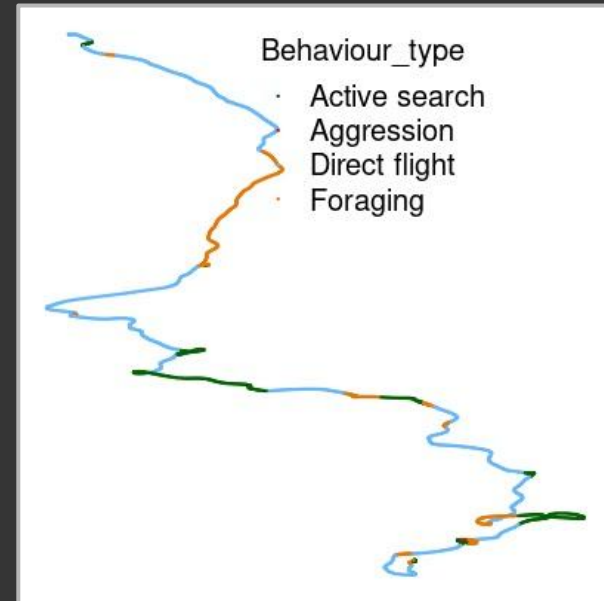
Assess effect of sampling interval, **dt**, on behavioural inference

- Specifying different **dt**
- Nyquist-Shannon sampling theorem to inform choice of **dt**



– Nyquist-Shannon sampling theorem (NSST)

Sampling at time interval **dt** is sufficient to correctly characterize behaviours that last **2 dt** or longer



— References

Zucchini, W., MacDonald, I. L. & Langrock, R. (2016). [Hidden Markov Models for Time Series: An Introduction Using R](#)

McClintock, B. T. & Michelot, T. (2018). [momentuhmm: R package for generalized hidden Markov models of animal movement](#), *Methods in Ecology and Evolution* 9 (6), 1518–1530

Nathan et al. (2022). [Big-data approaches lead to an increased understanding of the ecology of animal movement](#). *Science*. 375(6582). doi:10.1126/science.abg1780

Thanks to

Supervisory team



EPSRC

Engineering and Physical Sciences
Research Council



MAXWELL INSTITUTE FOR
MATHEMATICAL SCIENCES



JNCC
Joint Nature Conservation Committee



THE UNIVERSITY
of EDINBURGH

Thanks



for your attention